

1. Degenerate cases. If $a = 0$ then the right-hand side equals $\frac{b^q}{q} \geq 0$ while the left-hand side is 0; the inequality holds. If $b = 0$ the argument is symmetric. Hence we may assume $a > 0$ and $b > 0$, so that all logarithms below are well defined.

2. Weighted AM–GM via Jensen’s inequality. Consider the convex function $f(t) = e^t$ on $\mathbb{R}$ (since $f''(t) = e^t > 0$). Apply Jensen’s inequality to the two points

$$x_1 = \ln a^p, \quad x_2 = \ln b^q,$$

with weights

$$\lambda_1 = \frac{1}{p}, \quad \lambda_2 = \frac{1}{q}.$$

Because $\lambda_1, \lambda_2 \geq 0$ and $\lambda_1 + \lambda_2 = \frac{1}{p} + \frac{1}{q} = 1$, Jensen gives

$$f(\lambda_1 x_1 + \lambda_2 x_2) \leq \lambda_1 f(x_1) + \lambda_2 f(x_2). \quad (1)$$

Writing the weights explicitly we obtain

$$e^{\frac{1}{p} \ln a^p + \frac{1}{q} \ln b^q} \leq \frac{1}{p} e^{\ln a^p} + \frac{1}{q} e^{\ln b^q}.$$

3. Simplification. For any positive x we have $e^{\ln x} = x$. Moreover

$$e^{\frac{1}{p} \ln a^p} = a, \quad e^{\frac{1}{q} \ln b^q} = b,$$

since $\frac{1}{p} \ln a^p = \ln a$ and $\frac{1}{q} \ln b^q = \ln b$. Substituting these equalities into (2) yields

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}. \quad (3)$$

Thus the desired inequality holds for all $a, b > 0$. Together with the trivial cases from Step1, (3) is valid for all non-negative a, b .

4. Equality condition. For a strictly convex function, Jensen’s inequality is an equality ****iff**** the arguments are equal. Hence equality in (1) (and therefore in (3)) occurs exactly when

$$x_1 = x_2 \iff \ln a^p = \ln b^q \iff a^p = b^q.$$

When $a = b = 0$ both sides of (3) are zero, so equality also holds. No other choice of (a, b) gives equality. Consequently the equality case is precisely

$$a^p = b^q \quad (\text{including } a = b = 0).$$

□